

Syllabus

Course Information

Course Number: CSCE 633
Course Title: Machine Learning
Section: 700
Time: Asynchronous
Location: Online
Credit Hours: 3

Instructor Details

Instructor: Bobak Mortazavi
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Office Hours: By Appointment

Course Description

Machine learning is a sub-field of Artificial Intelligence that gives computers the ability to learn and/or act without being explicitly programmed. Applications of machine learning have permeated many aspects of every-day life and can be found among others in self-driving cars, speech recognition, computer vision, and genomics. Topics include supervised and unsupervised learning (including parametric and non-parametric models, clustering, dimension reduction, deep learning), optimization procedures, and statistical inference.

Course Prerequisites

MATH 304 and STAT 211, and (CSCE 221 or STAT 404) or equivalent.

Course Learning Outcomes

The objective of this course is to teach fundamental methods of machine learning with focus on the theoretical underpinnings, practical implementations, and experimentation. Upon completion of the course students will:

1. Have a good understanding of the fundamental issues and challenges of machine learning: data, model selection, model complexity, etc.
2. Gain an understanding of the strengths and weaknesses of many popular machine learning approaches.
3. Uncover the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and unsupervised learning.
4. Be able to design and implement various machine learning algorithms in a range of real-world applications.

Textbook and/or Resource Materials

Introduction to Statistical Learning with Applications in Python, James, Witten, Hastie, Tibshirani, and Taylor

Elements of Statistical Learning, 2nd Edition, Hastie, Tibshirani, and Friedman

Optional Additional Resources:

Introduction to Machine Learning, 4th edition, MIT Press by Ethem Alpaydin

Learning from Data, by Yaser S. Abu-Mostafa, Malik Magdon-Ismael, and Hsuan-Tien Lin

Machine Learning Refined, 2nd Edition, Watt, Borhani, and Katsaggelos

Machine Learning and Deep Learning using Python and Tensorflow, Konsani and Kadre

Grading Policy

- The cutoff for an 'A' will be at most 90% of total score, 80% for a 'B', 70% for a 'C', and 60% for a 'D'. Anything below 60% will be an 'F'.

1. Four homework assignments = 50%

2. One Final Project Assignment = 25%
3. Ten quizzes = 25%

Late Work Policy

- No Late Assignments will be accepted. Late assignments are any deliverables after the established deadline.

Work submitted by a student as makeup work for an excused absence is not considered late work and is exempted from the late work policy ([Student Rule 7 \(https://student-rules.tamu.edu/rule07/\)](https://student-rules.tamu.edu/rule07/)).

Course Schedule

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Course Topics and Major Assignment Dates

Module	Topic	Assignments, and Due Dates
1	Introduction and Math Review	Quiz 1, May 30
2	Linear Perceptron	Quiz 2, June 3
3	Linear Regression	Quiz 3, June 6
4	Logistic Regression & Regularization & Feature Selection	HW 1 Due, Quiz 4, June 10

5	Decision trees & random forest & boosting	HW 2 Due, Quiz 5, June 24
6	Support Vector Machines	HW 3 Due, Quiz 6, July 1
7	Neural Networks & Deep Learning	HW 4 Due, Quiz 7, July 15
8	Dimension Reduction	Quiz 8, Jul 22
9	Unsupervised Learning	Quiz 9, Jul 29
10	Advanced Topics	Quiz 10, Final Project Due August 5

Optional Course Information Items

1. Computer accounts: if you do not have a unix account, ask for one on the CS web page or register with TAMU HPRC
2. Programming languages permitted: Python and must be executable on CS unix hosts or other public systems in the department lab.
3. All written assignments must be implemented in Latex.

Use of Generative AI

This course assumes that all work submitted by students will be generated by the students themselves. Students should not have another person/entity do the writing of any substantive portion of an assignment for them, which includes hiring a person or a company to write assignments and using artificial intelligence tools like ChatGPT.

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to [Student Rule 7 \(https://student-rules.tamu.edu/rule07/\)](https://student-rules.tamu.edu/rule07/) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to [Student Rule 7 \(https://student-rules.tamu.edu/rule07/\)](https://student-rules.tamu.edu/rule07/) in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" ([Student Rule 7, Section 7.4.1 \(https://student-rules.tamu.edu/rule07/\)](https://student-rules.tamu.edu/rule07/)).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2 \(https://student-rules.tamu.edu/rule07/\)](https://student-rules.tamu.edu/rule07/)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. ([See Student Rule 24 \(https://student-rules.tamu.edu/rule24/\)](https://student-rules.tamu.edu/rule24/)).

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20 \(https://aggiehonor.tamu.edu/Rules-and-Procedures/Rules/Honor-System-Rules\)](https://aggiehonor.tamu.edu/Rules-and-Procedures/Rules/Honor-System-Rules)).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at [aggiehonor.tamu.edu \(https://aggiehonor.tamu.edu/\)](https://aggiehonor.tamu.edu/).

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Resources office on your campus (resources listed below). Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Disability Resources is located in the Student Services Building or at (979) 845-1637 or visit <https://disability.tamu.edu/> (<https://disability.tamu.edu/>).

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see [University Rule 08.01.01.M1](https://rules-saps.tamu.edu/PDFs/08.01.01.M1.pdf) (<https://rules-saps.tamu.edu/PDFs/08.01.01.M1.pdf>)):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, a person who is subjected to the alleged conduct will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns related to mental and/or physical health in a confidential setting are encouraged to make an appointment with [University Health Services](https://uhs.tamu.edu/appointments/index.html) (<https://uhs.tamu.edu/appointments/index.html>) or download the [TELUS Health Student Support app](https://uhs.tamu.edu/mental-health/student-support.html) (<https://uhs.tamu.edu/mental-health/student-support.html>) for 24/7 access to professional counseling in multiple languages. Walk-in services for urgent, non-emergency needs are available during normal business hours at University Health Services locations; call 979.458.4584 for details.

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's [Title IX webpage \(https://titleix.tamu.edu/\)](https://titleix.tamu.edu/).

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors influencing a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care practices by utilizing the resources and services available through [University Health Services \(https://uhs.tamu.edu/\)](https://uhs.tamu.edu/). Students needing a listening ear, information on resources, or an opportunity to build community can connect with an Aggie Peer Advocate through Aggie Peer Connect. Hours and locations can be found at University Health Services, [Aggie Peer Connect \(https://uhs.tamu.edu/prevention/aggie-peer-connect.html\)](https://uhs.tamu.edu/prevention/aggie-peer-connect.html). The [TELUS Health Student Support app \(https://uhs.tamu.edu/mental-health/student-support.html\)](https://uhs.tamu.edu/mental-health/student-support.html) provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or 988lifeline.org  [\(https://988lifeline.org/\)](https://988lifeline.org/).

Students needing a listening ear, information on resources, or an opportunity to build community can connect with an Aggie Peer Advocate through Aggie Peer Connect. Hours and locations can be found at University Health Services, [Aggie Peer Connect \(https://uhs.tamu.edu/prevention/aggie-peer-connect.html\)](https://uhs.tamu.edu/prevention/aggie-peer-connect.html). The [TELUS Health Student Support app \(https://uhs.tamu.edu/mental-health/student-support.html\)](https://uhs.tamu.edu/mental-health/student-support.html) provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or 988lifeline.org  [\(https://988lifeline.org/\)](https://988lifeline.org/).

